Netflix Data Analysis and Business Insights Report

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Executive Summary

The project involves an end-to-end analysis of a dataset from Netflix which contains details about the movies and television shows available on the platform. The dataset includes 8,790 titles and has attributes such as content type, title, director, country, date added, release year, rating, duration, and categories.

The analysis made use of Python, Pandas, NumPy, Matplotlib and Power BI. The project was concerned with gaining an understanding of Netflix's content distribution, the production countries, trends relating to the year of release, the various categories, the audience ratings, the lengths of movies and the number of seasons of TV shows.

The analysis found that movies were very prominent, there being 6,126 Movies (69.7%) as compared to 2,664 TV Shows (30.3%). The United States was the country identified as being most closely associated with Netflix content. Moreover, the analysis indicated a wide variety of genres, ratings, release years, movie durations, and TV-show seasons, showing the diversity of Netflix's content library.

The findings were turned into business insights and were displayed using an interactive Power BI dashboard called "NETFLIX BUSINESS INSIGHTS", which included KPI cards, an analysis of content type, trends relating to the release year, an analysis of the top countries, an analysis by category, a distribution of ratings, and interactive slicers.

The project as a whole shows a full data-analysis workflow, beginning with the inspection and preparation of the raw data and then moving on to exploratory analysis, visualization, business interpretation, and the development of an interactive dashboard.

1. Introduction

1.1 Background

Netflix is a huge global service which provides a wide and varied selection of films and television programmes. Since there is an ever growing amount and range of content available through streaming, there is the possibility of using data analysis in order to understand the features of the content library.

An analysis of Netflix's catalogue should help to reveal patterns relating to content types, the years in which they were released, the countries of production, the categories, the ratings, the durations of movies, and the number of seasons in television shows. Such insights could contribute to a deeper understanding of Netflix's content strategy and serve as a practical illustration of how data analysis can be carried out on a real-world dataset.

The project makes use of Python and Power BI to analyse Netflix's content data with the aim of identifying significant patterns and turning them into valuable business insights.

1.2 Problem Statement

The dataset that Netflix has contains a number of attributes relating to thousands of films and television programmes. Yet the raw data by itself does not make clear important patterns such as:

- The proportion of movies and TV shows
- Changes in content across release years
- Countries contributing the most content
- Popular content categories
- Distribution of audience ratings
- Typical movie durations
- Number of seasons in TV shows

The kind of problem that this project sets out to deal with is to turn the original Netflix dataset into useful analytical information and to create an interactive dashboard which makes it easier to understand these patterns.

1.3 Objectives

The main objectives of this project were to:

1. Look over and fully understand the Netflix dataset.
2. Identify and deal with missing and duplicate records.
3. Handle complex fields such as those relating to country and category.
4. Look at movies and TV shows individually.
5. Look at how Netflix distributes its content according to the year of release.
6. Carry out an analysis of the content by country.
7. Find the main categories and types.
8. Analyze rating distribution.
9. Look at the length of movies and the number of seasons in TV shows.
10. Provide business insights that can be put into action.
11. Create an interactive Power BI dashboard.
12. Put the results in a clear and professional format.

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2. Dataset Overview

The information relating to the movies and TV shows available on Netflix is included in the Netflix dataset.

The dataset contains the following columns:

Column	Description
show_id	Unique identifier of the title
type	Indicates whether the title is a Movie or TV Show
title	Name of the movie or TV show
director	Director associated with the title
country	Country or countries associated with the title
date_added	Date when the title was added to Netflix
release_year	Original release year of the title
rating	Audience/content rating
duration	Movie duration or number of TV-show seasons
listed_in	Categories or genres associated with the title

After preparation and analysis the dataset has 8,790 titles.

The main content distribution is:

- Movies: 6,126
- TV Shows: 2,664
- Total: 8,790

Movies account for about 69.7% of the dataset, and TV Shows for about 30.3%.

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3. Tools and Technologies

The project used a combination of Python based data analysis tools and Power BI to prepare, analyze, visualize, and present the Netflix dataset.

Python

Python was the main programming language used throughout the project. It was used for data preparation, analysis, transformation, and visualization.

Pandas

Pandas was one of the primary libraries used for working with the dataset. It was used for:

- Loading the Netflix dataset
- Inspecting the data
- Checking missing values and duplicates
- Filtering records
- Transforming data

- Grouping and aggregating information
- Analyzing countries and content categories

NumPy

NumPy was used for numerical operations and supported different data-processing tasks during the analysis.

Matplotlib

Matplotlib was used to create charts and visualizations. These visualizations helped make trends, distributions, and comparisons within the dataset easier to understand.

Google Colab / Jupyter Environment

The Python analysis was carried out in a notebook environment. This made it possible to keep data preparation, analysis, visualization, and documentation together in one workflow.

Power BI

Power BI was used to create the final interactive dashboard. The main findings were presented through KPI cards, charts, and interactive slicers, making the results easier to explore and understand.

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4. Cleaning and Preparation of the Data

Data preparation was an important part of the project because some fields in the original dataset required additional processing before they could be used effectively for analysis.

4.1 Data Inspection

The dataset was first inspected to understand its overall structure and quality. The inspection focused on:

- Number of rows and columns
- Column names
- Data types
- Missing values
- Duplicate records
- Unique values
- Distribution of major categorical fields

The dataset contained the following columns:

show_id, type, title, director, country, date_added, release_year, rating, duration, and listed_in

The inspection showed that the dataset contained a combination of categorical, numerical, and date-related information, making it suitable for exploratory data analysis.

4.2 Missing Values

Missing values were checked across the dataset. Some fields, particularly director, country, and other descriptive attributes, contained missing information.

Rather than treating missing values as meaningful categories, they were handled according to the requirements of each analysis. When a particular field was required for an analysis, records with unavailable information were either excluded from that calculation or considered separately.

This helped prevent missing information from affecting the results or creating misleading conclusions.

4.3 Duplicate Records

The dataset was also checked for duplicate records to ensure that repeated entries did not affect content counts.

The show_id field was used as an important reference when checking the uniqueness of records. Where duplicate records were identified, they were handled appropriately so that the analysis represented actual content records rather than repeated entries.

4.4 Date Processing

The date_added field represents the date on which content was added to Netflix.

The field was reviewed and processed so that it could be used appropriately for analysis. Date information is particularly useful for understanding how Netflix's content additions change over time and can also support future time-based analysis.

4.5 Country Processing

The country column required additional attention because a single title can be associated with more than one country.

For example, one title may have multiple countries listed in the same field. Simply counting the original values would therefore make country level comparison less accurate.

The country information was separated and cleaned so that individual countries could be analyzed independently. This made it possible to calculate content associations by country and identify the countries with the strongest representation in the Netflix catalog.

4.6 Category Processing

The listed_in column contains one or more categories associated with each title.

To analyze individual genres and categories, the combined category information was separated and processed. This allowed the project to identify frequently represented categories and examine the overall variety of Netflix content.

4.7 Movie Duration Processing

The duration column contains different types of information depending on the content type.

For movies, duration is recorded in minutes. This information was extracted and converted into a numerical Duration_Minutes value.

This transformation made it possible to perform numerical analysis and create visualizations showing the distribution of movie lengths.

4.8 TV Show Season Processing

For TV Shows, the duration field represents the number of seasons rather than minutes.

The season information was therefore extracted into a separate numerical field. This allowed the number of seasons to be analyzed independently from movie duration.

Keeping these two measurements separate prevented movies and TV shows from being incorrectly compared using the same duration metric.

4.9 Data Type Conversion

The dataset was reviewed to ensure that the fields were in suitable formats for analysis.

The relevant fields were already in appropriate formats, including:

- Text fields for identifiers and categorical information
- Numerical values for release_year
- Date information for date_added
- Text fields for ratings and duration before specialized processing

As a result, extensive data-type conversion was not necessary. The main preparation work focused on verifying the existing formats and transforming fields such as duration, country, and category information where required.

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5. Exploratory Data Analysis

Exploratory Data Analysis (EDA) was carried out to understand the structure of the Netflix catalog and identify important patterns within the data.

The main areas examined were:

- Movies vs TV Shows
- Release years
- Countries

- Categories and genres
- Ratings
- Movie durations
- TV Show seasons

5.1 Movies vs TV Shows

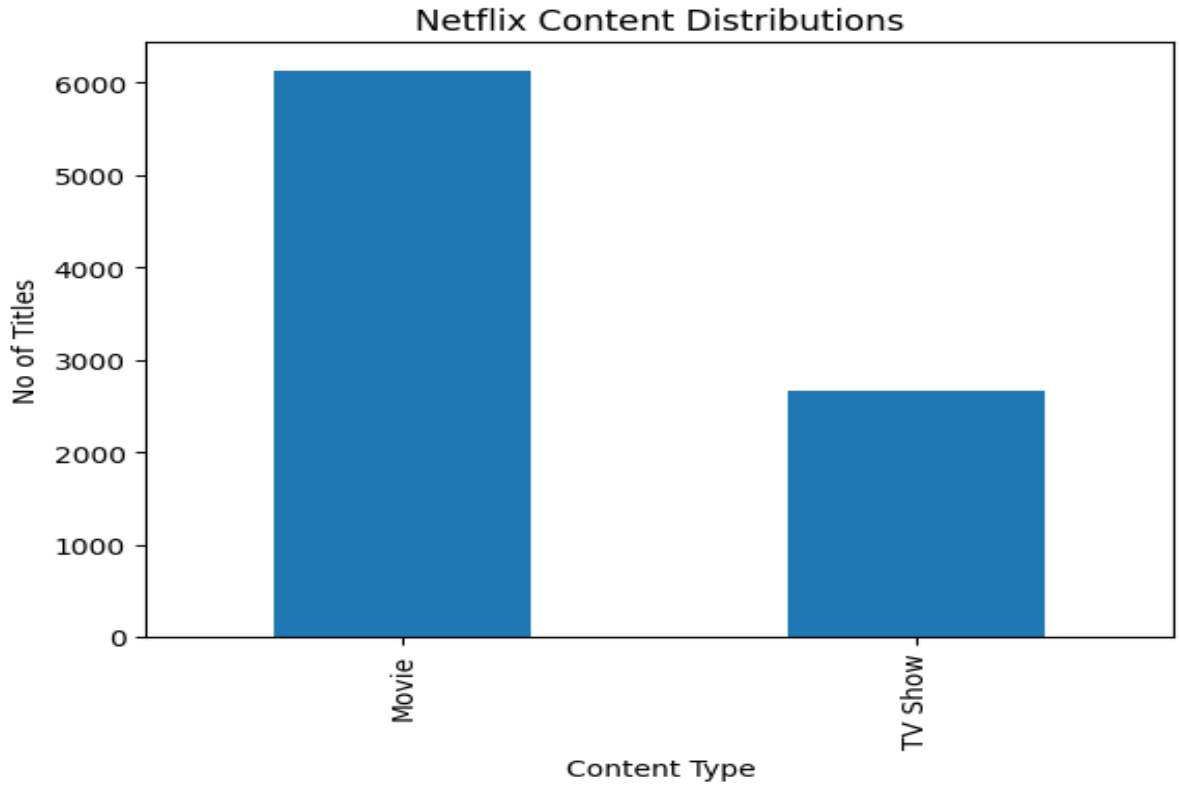
The analysis shows that movies make up the majority of the Netflix content included in the dataset.

Content Type	Number of Titles	Percentage
Movies	6,126	69.7%
TV Shows	2,664	30.3%
Total	8,790	100%

Movies account for approximately 69.7% of the dataset, while TV Shows account for approximately 30.3%.

This means that the Netflix catalog analyzed in this project has a considerably larger representation of movies than TV shows.

The difference was also presented visually through the content type analysis in Python and the Power BI dashboard.

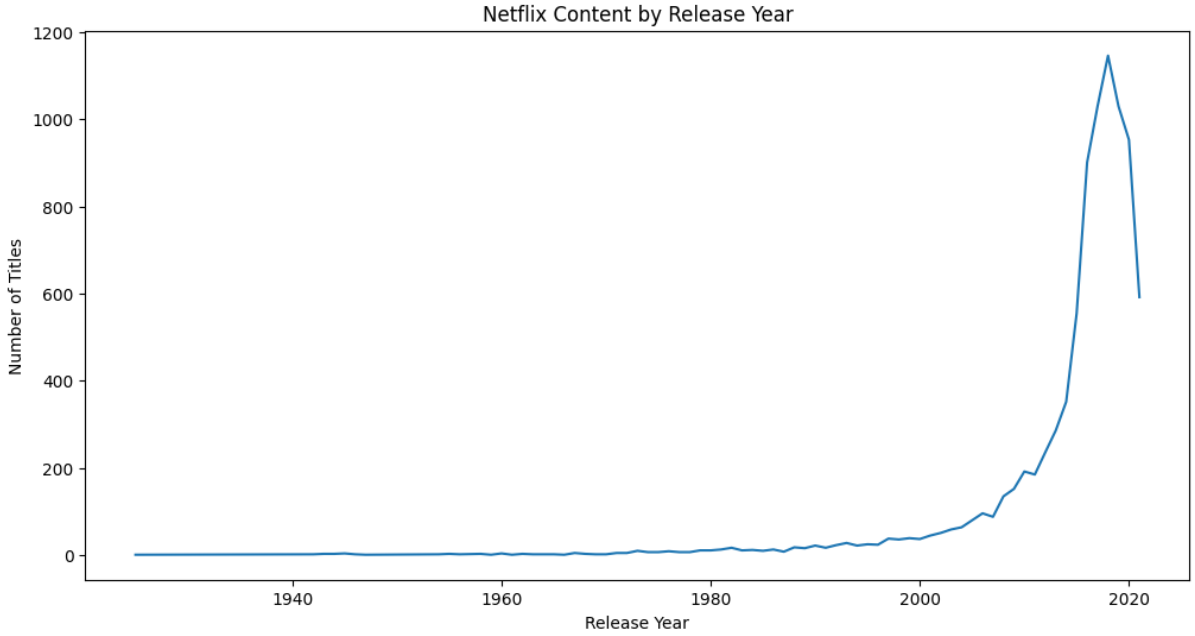


5.2 Release Year Analysis

The release_year field was analyzed to understand how Netflix titles are distributed across different release periods.

The dataset covers a broad range of release years, with a strong representation of relatively recent titles. The release year visualization makes it easier to see how the number of titles changes over time and which periods contain higher concentrations of content.

The distribution suggests that relatively recent content represents an important part of the Netflix catalog analyzed in this project.



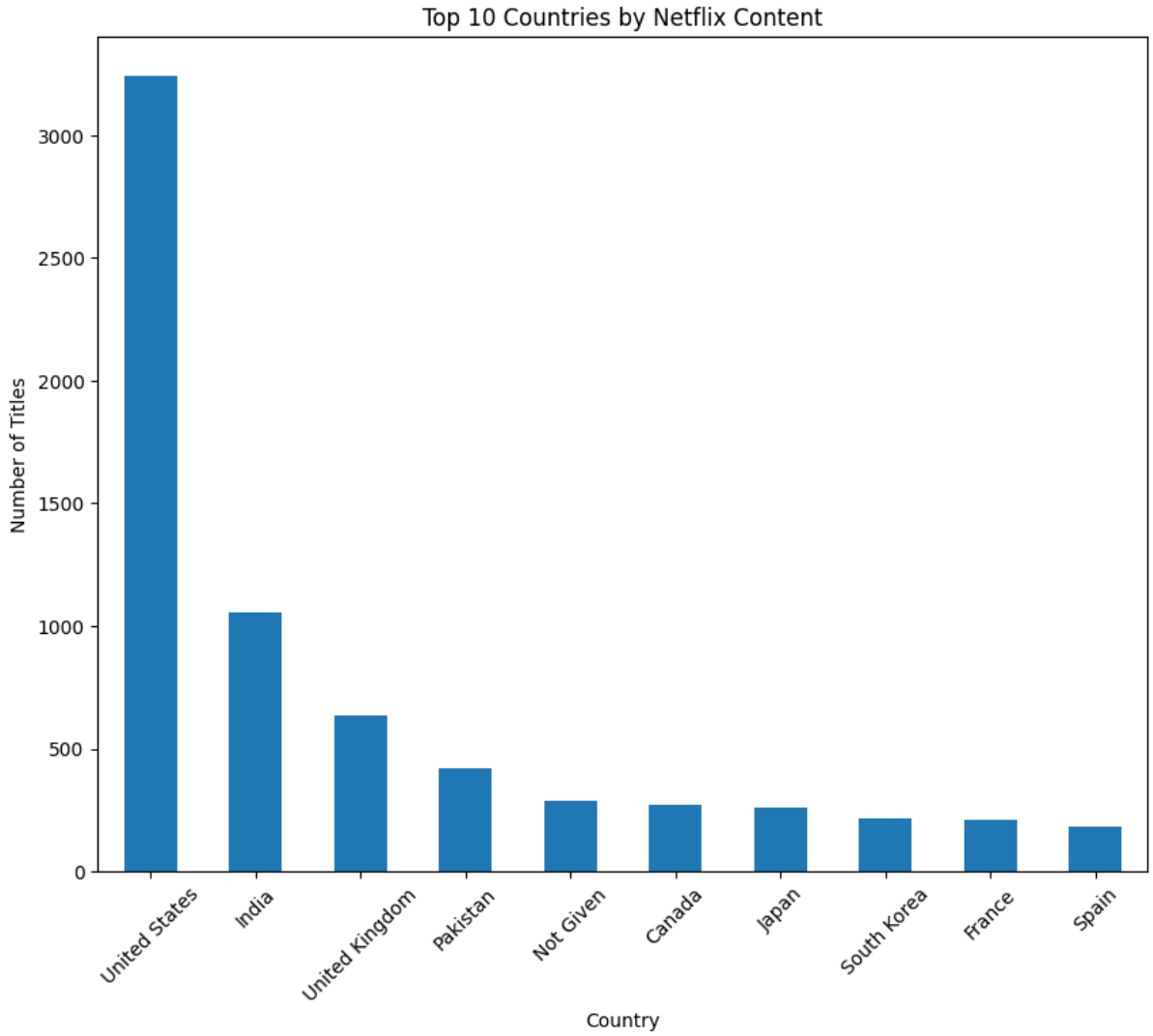
5.3 Country Wise Content Analysis

Country wise analysis was performed by separating titles associated with multiple countries and counting each country independently.

The United States emerged as the leading country associated with Netflix content in the dataset. This highlights the strong presence of U.S associated productions within the catalog.

At the same time, the presence of content associated with many other countries shows that Netflix's catalog has a broad international reach.

The Top 10 Countries analysis was also included in the Power BI dashboard to make comparison between the leading countries easier.



5.4 Category / Genre Analysis

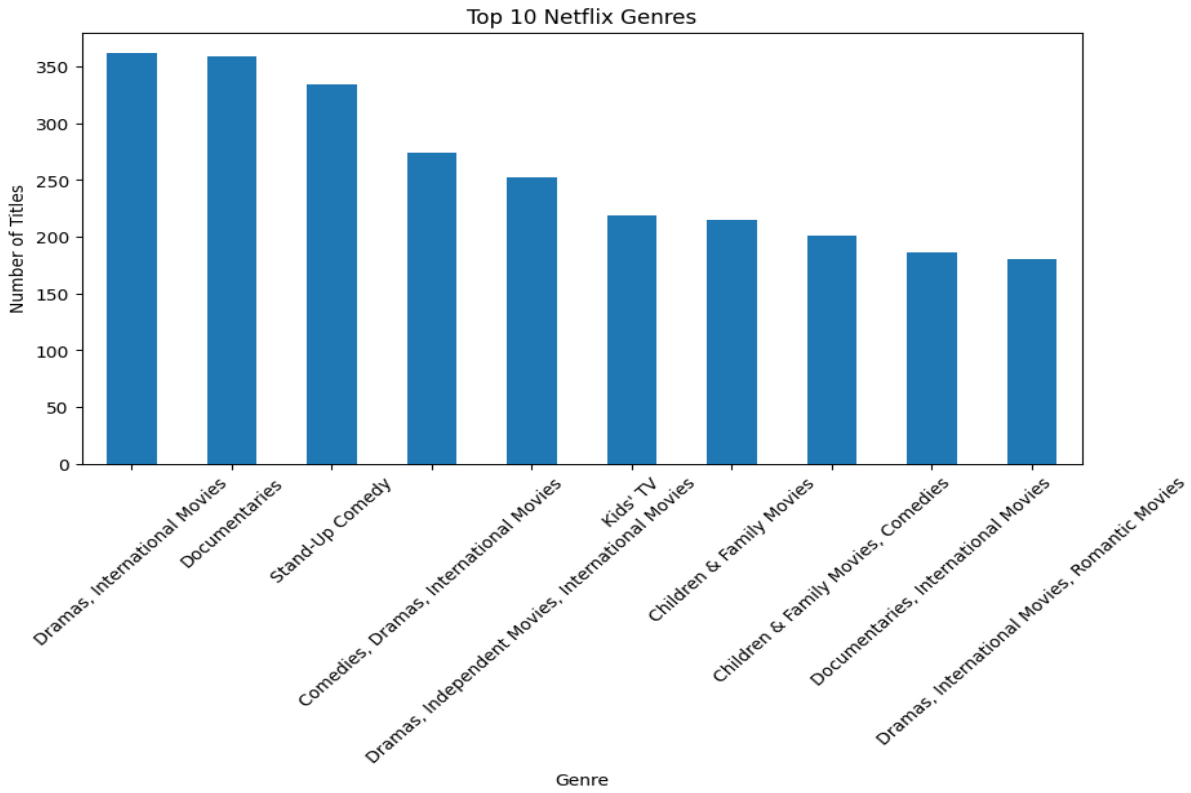
The listed_in column was analyzed to identify the main categories and genres represented in the Netflix catalog.

Since a single title can belong to multiple categories, the categories were separated before counting them. This provided a clearer picture of how frequently individual genres and categories appear in the dataset.

The results show a wide variety of content, including international content, dramas, comedies, documentaries, action oriented titles, and television related categories.

The Top Categories visualization makes it easier to compare the categories with the highest representation.

Overall, the range of categories shows that Netflix caters to audiences with different entertainment interests.

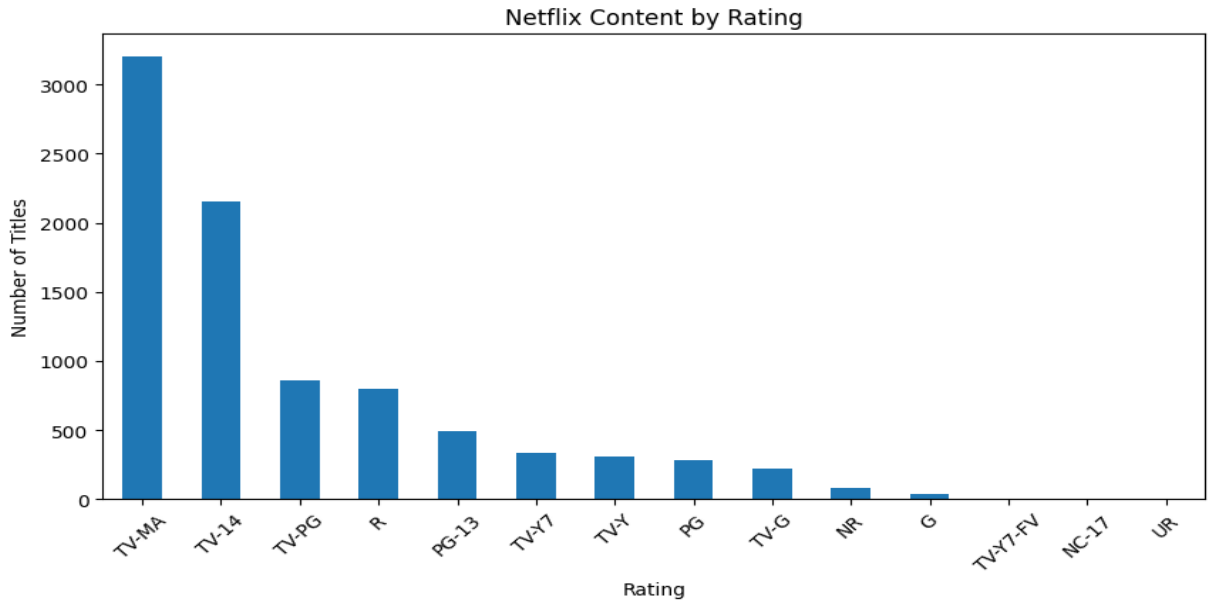


5.5 Rating Analysis

The rating column was analyzed to understand how Netflix content is distributed across different audience and content ratings.

The dataset contains ratings covering different audience groups, from general and family-oriented content to content intended for more mature audiences.

The Power BI dashboard presents this information through a rating distribution chart, allowing users to compare the number of titles associated with different ratings and get a clearer view of the audience coverage within the catalog.



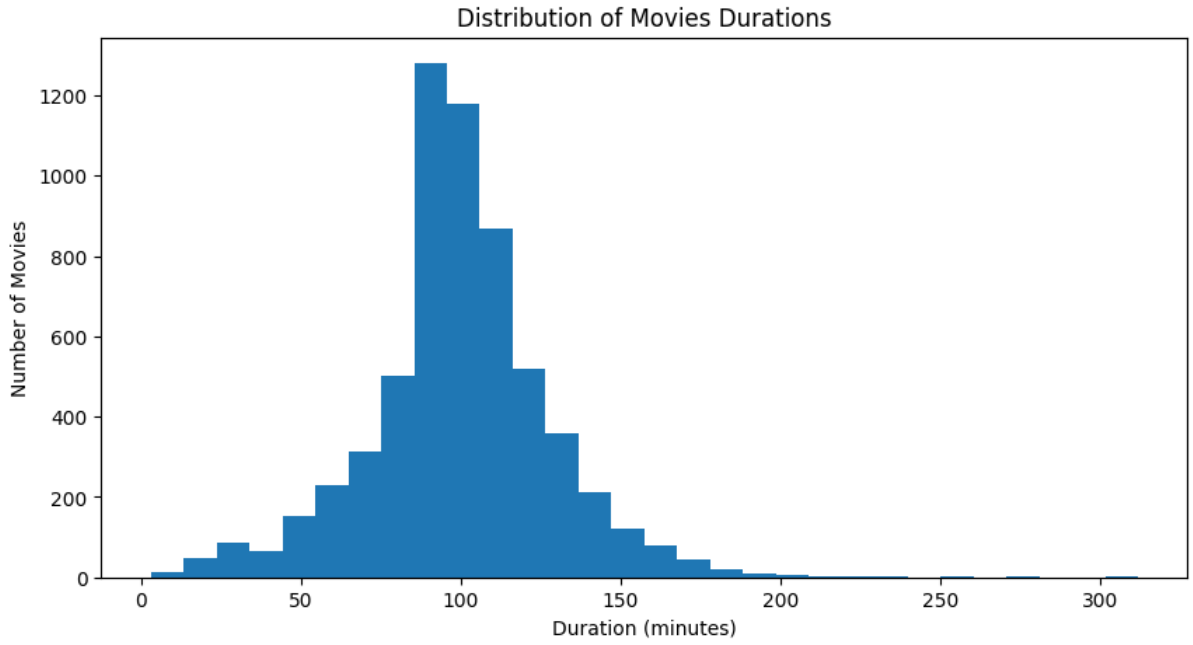
5.6 Movie Duration Analysis

Movie duration was extracted from the duration field and converted into a numerical value measured in minutes.

The distribution of movie lengths was then examined using statistical analysis and visualization.

The results show considerable variation in movie durations, meaning that the Netflix catalog contains both shorter and longer movies.

This analysis also highlights the importance of transforming the original duration field before using it for numerical analysis.



5.7 TV Show Season Analysis

For TV Shows, the duration field represents the number of seasons rather than minutes.

The season information was therefore extracted separately and analyzed as a numerical variable.

The results show that Netflix includes both shorter series and shows that continue across several seasons.

Separating the season information from movie duration ensured that each content type was analyzed using an appropriate measurement.

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6. Business Insights

The findings from the exploratory analysis were translated into business oriented observations that provide a clearer understanding of the Netflix catalog.

6.1 Strong Movie Presence

Movies account for 6,126 of the 8,790 titles, representing approximately 69.7% of the analyzed catalog.

This makes movies the dominant content format in the dataset and shows that they play a major role in the overall composition of the catalog.

6.2 Importance of Recent Content

The release year analysis shows a strong representation of relatively recent titles.

This highlights the importance of newer content within the Netflix catalog and suggests that maintaining a regularly updated collection is an important part of Netflix's content strategy.

6.3 Strong U.S Content Representation

The United States is the leading country associated with Netflix content in the analysis.

This reflects the significant role of U.S associated productions within the catalog.

6.4 International Content Diversity

Although the United States has the highest representation, the dataset also contains content associated with many other countries.

This highlights the international nature of Netflix's content library and its ability to offer content from different production markets.

6.5 Diverse Content Categories

The category analysis shows a wide range of genres and content categories.

This variety allows Netflix to serve users with different entertainment preferences instead of relying on a single type of content.

6.6 Broad Audience Coverage

The rating distribution includes multiple audience categories.

This suggests that the catalog serves different audience groups rather than being concentrated around one specific demographic.

6.7 Variety in Content Length

The movie duration analysis shows variation in movie lengths, while TV shows differ in the number of seasons.

This means the catalog offers different viewing experiences, ranging from shorter movies and limited series to longer-running content.

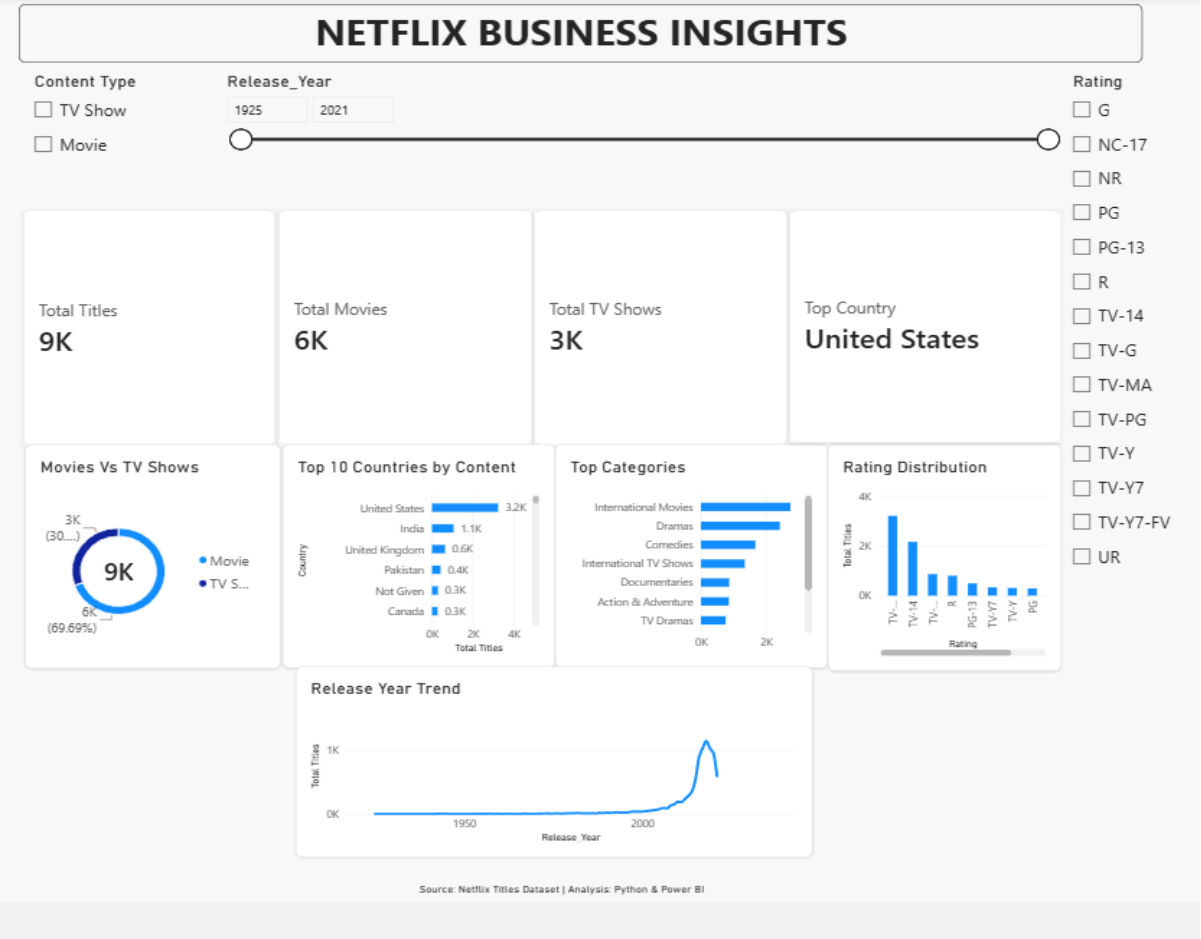
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7. Power BI Dashboard

The final stage of the project was the development of an interactive Power BI dashboard titled:

NETFLIX BUSINESS INSIGHTS

The dashboard brings together the main findings from the analysis in a single visual interface, allowing users to understand the key results quickly and explore the data interactively.



7.1 Dashboard Title

The dashboard is titled NETFLIX BUSINESS INSIGHTS.

The title clearly identifies the purpose of the dashboard and provides a professional heading for the visual analysis.

7.2 KPI Cards

Four main KPI cards were included in the dashboard:

Total Titles — 8,790

Shows the total number of Netflix titles included in the analysis.

Movies — 6,126

Shows the total number of movies in the dataset.

TV Shows — 2,664

Shows the total number of TV shows.

Top Country — United States

Shows the leading country associated with Netflix content in the analysis.

Together, these KPI cards provide a quick overview of the most important figures before users explore the detailed charts.

7.3 Movies vs TV Shows

A visual comparison between Movies and TV Shows was included to show the overall composition of the Netflix catalog.

The chart clearly shows that movies make up approximately 69.7% of the titles, compared with 30.3% for TV shows.

This allows users to understand the content structure at a glance.

7.4 Release Year Trend

A release-year visualization was included to show how Netflix titles are distributed across different years.

The chart allows users to identify periods with higher concentrations of titles and observe how the content distribution changes over time.

It also supports the finding that relatively recent content represents an important part of the catalog.

7.5 Top 10 Countries

The dashboard includes a Top 10 Countries visualization that ranks countries based on their associated Netflix titles.

The United States appears as the leading country, while the remaining countries highlight the international nature of the catalog.

Presenting this information visually makes geographic comparison much easier than working with the raw country data.

7.6 Top Categories

The Top Categories chart presents the most frequently represented categories in the dataset.

This helps users quickly identify the major content categories and provides a visual representation of the diversity within Netflix's catalog.

7.7 Ratings Distribution

A ratings distribution chart was included to compare the number of titles associated with different ratings.

This provides a quick overview of the types of audiences represented in the Netflix catalog.

7.8 Interactive Slicers

Interactive slicers were added to make the dashboard easier to explore.

The main slicers include:

- Content Type
- Release Year
- Rating

Users can select Movies or TV Shows, choose a particular release year, or focus on a specific rating. The charts and KPI information then respond to the selected filters, allowing users to examine specific parts of the dataset.

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8. Dashboard Design and Usability

The dashboard was designed to present the most important information clearly while allowing users to explore the data without interacting directly with the underlying dataset.

The layout follows a logical information hierarchy:

1. Dashboard title
2. KPI cards
3. Overall content distribution
4. Time-based analysis
5. Geographic analysis
6. Category analysis
7. Rating analysis
8. Interactive filtering

The KPI cards at the top provide an immediate summary of the major figures, while the charts provide more detailed information.

The interactive slicers further improve usability by allowing users to filter the dashboard according to content type, release year, or rating.

As a result, the dashboard works both as a summary of the analysis and as an interactive business intelligence tool.

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9. Methodology

The project followed a structured data analysis workflow from data collection through dashboard development.

Step 1: Data Collection

The Netflix dataset was obtained and loaded into the Python analysis environment.

Step 2: Data Inspection

The dataset structure, column names, data types, missing values, and duplicate records were examined.

Step 3: Data Cleaning

Missing values and duplicate records were investigated and handled according to the requirements of the analysis.

Step 4: Data Transformation

Complex fields such as:

- Country
- Categories
- Movie duration
- TV Show seasons

were transformed into formats that could be analyzed more effectively.

Step 5: Exploratory Data Analysis

Python was used to investigate:

- Content types
- Release years
- Countries
- Categories
- Ratings
- Movie duration
- TV Show seasons

Step 6: Visualization

Charts and graphs were created to make important patterns and comparisons easier to understand.

Step 7: Business Insights

The analytical results were interpreted from a business perspective to identify useful observations about Netflix's content catalog.

Step 8: Power BI Dashboard

The prepared data and major findings were presented through an interactive Power BI dashboard.

Step 9: Dashboard Interpretation

The final dashboard was reviewed to ensure that the main findings were clearly represented and that users could explore the information through interactive filters.

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10. Limitations

Although the project provides useful insights into Netflix's content catalog, there are several limitations that should be considered.

Dataset Scope

The analysis is based on the available Netflix dataset. Therefore, it represents the information contained within that dataset rather than Netflix's complete and continuously changing catalog.

Missing Information

Some records contain missing values, particularly in fields such as director and country. This limits certain types of analysis.

Country Attribution

A title can be associated with multiple countries. Therefore, country counts represent country associations and should not necessarily be interpreted as exclusive production by a single country.

Category Overlap

A title can belong to multiple categories. As a result, category counts represent category associations rather than mutually exclusive groups.

No Viewer Level Data

The dataset does not contain individual user information such as:

- Watch history
- Viewing frequency

- Watch time
- User ratings
- Completion rates
- User demographics

Because of this, the project focuses on content characteristics rather than actual viewer behavior.

Descriptive Nature of the Analysis

The project mainly focuses on descriptive and exploratory analysis. It does not predict future content performance or individual user preferences.

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11. Future Scope

The project can be extended in several ways if additional data and resources become available.

11.1 Viewer Behaviour Analysis

If user level viewing data were available, future analysis could examine:

- Most watched genres
- Viewing frequency
- Average watch time
- Completion rates
- Viewer preferences
- Content engagement

This would allow the project to move beyond content analysis and explore actual user behavior.

11.2 Recommendation System

A recommendation system could be developed using features such as:

- Content categories
- Ratings
- Country
- Release year
- Movie/TV Show type
- Similarity between titles

Machine learning techniques could then be used to recommend content based on user preferences or similarities between titles.

11.3 Predictive Analysis

With suitable historical target variables, machine-learning models could be developed to predict different content related outcomes.

Possible applications include:

- Content popularity prediction
- Genre popularity forecasting
- Rating prediction
- Content demand forecasting

11.4 Regional Analysis

The country level analysis could be expanded into a deeper regional study.

For example, content could be compared across:

- North America
- Europe
- Asia
- Latin America
- Africa
- Other regions

This could provide additional insight into Netflix's international content strategy.

11.5 Advanced Power BI Dashboard

The dashboard could also be expanded with features such as:

- Detailed drill through pages
- Geographic maps
- Advanced DAX measures
- Year over Year comparisons
- Dynamic KPI calculations
- Additional filters
- Custom tooltips
- Bookmark based navigation
- Dedicated country and genre analysis pages

These additions could turn the current dashboard into a more comprehensive business intelligence solution.

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12. Conclusion

This project demonstrated a complete end-to-end data analysis workflow using a Netflix content dataset.

The work began with dataset inspection and preparation and continued through data cleaning, transformation, exploratory analysis, visualization, business interpretation, and Power BI dashboard development.

The analysis identified 8,790 Netflix titles, including 6,126 Movies and 2,664 TV Shows. Movies account for approximately 69.7% of the analyzed catalog, while TV Shows account for approximately 30.3%.

The analysis also identified the United States as the leading country associated with Netflix content and highlighted the diversity of the catalog in terms of categories, ratings, release years, movie durations, and TV-show seasons.

The final NETFLIX BUSINESS INSIGHTS Power BI dashboard brings these findings together through KPI cards, Movies vs TV Shows analysis, release-year trends, Top 10 Countries, Top Categories, Ratings Distribution, and interactive slicers.

Overall, the project provided practical experience with Python, Pandas, NumPy, data preparation, exploratory data analysis, visualization, business insight generation, and Power BI dashboard development. More importantly, it demonstrated how a raw dataset can be transformed into meaningful information that can support analysis and decision making.

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13. Final Project Outcome

The final outcome of the project is a complete Netflix Data Analysis solution covering the full journey from raw data to an interactive business intelligence dashboard.

Data Analysis

A structured Python based analysis of the Netflix dataset covering content types, release years, countries, categories, ratings, movie durations, and TV-show seasons.

Data Preparation

A prepared and transformed dataset suitable for analysis and visualization.

Exploratory Analysis

Multiple analyses and visualizations were used to identify important patterns and characteristics within the Netflix catalog.

Business Insights

The project produced business-oriented findings related to movie dominance, recent content, U.S content representation, international diversity, category diversity, audience coverage, and content length.

Power BI Dashboard

The interactive NETFLIX BUSINESS INSIGHTS dashboard includes:

- 8,790 Total Titles KPI
- 6,126 Movies KPI
- 2,664 TV Shows KPI
- United States Top Country KPI
- Movies vs TV Shows visualization
- Release Year Trend
- Top 10 Countries
- Top Categories
- Ratings Distribution
- Content Type slicer
- Release Year slicer
- Rating slicer

Overall Achievement

The project successfully transformed a raw Netflix dataset into a structured analytical solution and interactive business intelligence dashboard.

The complete workflow can be summarized as:

Raw Data → Data Preparation → Exploratory Analysis → Visualization → Business Insights → Power BI Dashboard

It demonstrates practical experience in Python, Pandas, NumPy, Matplotlib, and Power BI, while showing how raw data can be transformed into meaningful insights for decision making.
